

Batteries

Seaglider's main battery pack is not just an energy store – it **is the trim mechanism**. The large main battery, with a brass weight bolted to its underside, is the mass that the mass shifter slides fore/aft for pitch and rotates for roll. That double duty means every battery change touches almost everything else: vehicle mass, ballast, pitch/roll trim, and even the compass calibration.



Source

Paraphrased from the APL-UW IOP *SGX Documentation* (v1.0, 2024), the Electrochem primary-lithium *Safety and Handling Guidelines*, and the battery MSDS/SDS sheets (UN3090/UN3091), and 2017–2024 correspondence between Seaglider operators, APL-UW IOP, the manufacturer and service providers (launch refusals, voltage cutoffs at sea, early pack exhaustion, battery changes). Pack configurations vary by vehicle generation and build – defer to APL-UW IOP and your battery vendor's documentation.

Pack architectures by generation

Generation	Packs	Notes
Legacy SG (1KA era)	One 24 V + one 10 V lithium primary	24 V pack drives the motors (VBD, mass shifter) and is the moving trim mass; 10 V pack lives under the main electronics board and powers electronics
SG (univolt)	15 V packs, 12.0 kg total, 0.56 kg lithium	Effective capacity ~350 Ah; the 10 V/24 V distinction disappears (but voltage-select jumpers must still be installed)
SGX	Three 15 V packs, 19.6 kg total, 0.91 kg lithium	Effective capacity ~575 Ah; larger forward pack plus an additional aft pack behind the mass shifter